

Operators :

-->Anything which can perform an operation is called as Operator

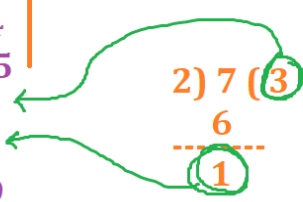
-->In Python we have different types of Operators, they are

- | | |
|-----------------------------------|--------------------------------|
| 1. Arithmetic Operators | : +, -, *, /, //, %, ** |
| 2. Relational Operators | : <, <=, >, >=, ==, != |
| 3. Logical Operators | : and, or, not |
| 4. Assignment Operators | : = |
| 5. ShortHand Assignment Operators | : +=, -=, *=, /=, //=, %=, **= |
| 6. Membership Operators | : in, not in |

1. Arithmetic Operators :

-->Arithmetic Operators are used to perform calculations

Operator	Meaning	Ex
+	Addition	7 + 2 --> 9
-	Subtraction	7 - 2 --> 5
*	Multiplication	7 * 2 --> 14
/	Division	7 / 2 --> 3.5
//	Floor Division	7 // 2 --> 3
%	Modulo Division	7 % 2 --> 1
**	Exponent/Power	7 ** 2 --> 49



--> How to Remove last digit in a given number

Ex: input : 125 n = 125
Output: 12 a = n // 10
 print(a)

```
10) 125 ( 12
    120
    ----
       5
```

--> How to get last digit from a given number

Ex: input : 125 n = 125
Output: 5 a = n % 10
 print(a)

2. Relational Operators

--> If you want to compare values then use Relational Operators

--> Relational Operators always gives boolean value as a result

--> if you want write conditions then use Relational Operators

Ex: n = int(input('Enter any Number : '))

n
5

n > 5	--> 5 > 5	--> False
n >= 5	--> 5 >= 5	--> True
n < 5	--> 5 < 5	--> False
n <= 5	--> True	
n == 5	--> True	
n != 5	--> False	

--> W.a.condition to check customer is eligible for vote or not

`age = int(input('Enter your Age : '))`

Condition: `age >= 18`

True
Eligible

False
Not Eligible

..., -3, -2, -1, **0**, 1, 2, 3,

--> W.a. condition to check given number is +Ve number or not

`n = int(input('Enter any Number : '))`

Condition: `n > 0`

--> W.a. condition to check Given number is Divisible by 5 or not

5) 8 (1
 5

 3

5) 15 (3
 15

 0

5) 22 (4
 20

 2

5) 30 (6
 30

 0

`n = int(input('Enter any Number : '))`

Condition: `n%5 == 0`

Even Number :

-->Any number which is divisible by 2 is called as Even Number

Ex: 2, 4, 6, 8, 10,

-->W.a.condition to check given number is Even or odd Number

`n = int(input('Enter any Number : '))`

Condition: `n%2 == 0`

-->W.a.condition to check customer is Eligible for discount or not

If customer bill Amount is 5000+ then eligible for discount

`billAmt = float(input('Enter Bill Amount '))`

Condition: `billAmt >= 5000`

Task:

1. W.a.c to check if a record ID is greater than 1000 before inserting into DB
2. W.a.c to verify if data batch size is equal to expected size (500 rows).
3. W.a.c to check if file size is less than 1MB, then process it.
4. W.a.c to check if number of failed records is not equal to 0.
5. W.a.c to verify if data pipeline execution time exceeds 10 minutes.
6. W.a.c to identify if customer rating is less than 3.
7. W.a.c to verify if ETL job status is not equal to 'Success'.

3. Logical Operators

--> If you want to combine morethan one condition results then use logical operators

--> If you want to check morethan one condition then use logical operators along with relational Operators

--> It always gives boolean value as a result

and --> If all condition results are True then only it return True

or --> atleast one condition is True then it return True

not --> It reverse given result That means True becomes False, False becomes True

Note:

Use AND when ALL conditions must be TRUE

Ex: Person must be 18+ AND salary > 20k

Use OR when ANY ONE condition is TRUE

Ex: if city == "Hyderabad" or city == "Bhubaneswar"

At least one is enough then accepted

Use NOT to reverse a condition

Ex: if not status == "Success":

Means → status is NOT success then do some operation

-->W.a.condition to check student is PASS or FAIL

s1 = int(input('Enter Sub-1 Marks : '))

s2 = int(input('Enter Sub-2 Marks : '))

s3 = int(input('Enter Sub-3 Marks : '))

Condition: s1>=35 and s2>=35 and s3>=35

--> W.a.condition to check person is eligible to apply job or not

min age is 18, Max age is 24 --->then only Eligible

age = int(input('Enter your age : '))

Condition: age>=18 and age<=24

-->W.a.condition to check customer is Eligible for discount or not

if customer has either SBI or ICICI card then eligible for discount

card = input('Enter your card : ')

Condition: card == 'SBI' or card == 'ICICI'

-->W.a.condition to check given character is in lower case or not

ch = input('Enter any Character : ')

Condition: ch>='a' and ch<='z'

Task:

--> W.a.condition to check Customer is Eligible for Half ticket or not
min age 5years max age 11 year

--> W.a.condition to check user enter invalid marks or not
Except 0 to 100 it is Invalid

--> W.a.condition to check given character is digit or not

➔ I have different files; I need to collect all files whose file size is min 1MB as well as record count is 1000+

➔ I have data of different persons and different countries, but i want to collect all Indian person's data whose age is 18+

➔ I have India wise customer data, data is divided by region wise like north, south, east and west. now i need collect all records whose region is either South or North

4. Assignment Operator(=)

--> '=' is used to assign values to variables

--> It is also called as Right to Left Operator

Syntax: $\text{Variablename} = \text{Value}$
 Left Right

5. ShortHand Assignment Operators

--> It is a combination of Arithmetic and Assignment Operators

--> It is used to reduce expression length

Ex: a = 5	c = 0	p = 5
b = 7	c = c + 1	q = 3
a = a + b	(or)	q = p * q
(or)		(or)
a += b	c += 1	q *= p

6. Membership Operators

--> in, not in are called as membership operators

--> To check your value available in a group of values or not then use membership operators

Ex: courses = ['Python', 'DB', 'Data Analysis', 'Power BI', 'Excel', 'Django']
c = input('Enter your course : ')
print(c in courses)

--> Check email is valid or not

```
email = "avd@gmail.com"  
if "@" in email:  
    print("Valid Email")
```